

UNIVERSITY OF BIRMINGHAM

School of Mathematics

Programmes in the School of Mathematics

Programmes involving Mathematics

Second Examination

Second Examination

2LA 06 15552 Level I

Linear Algebra

January Examinations 2022-23

One Hour and Thirty Minutes

Full marks will be obtained with complete answers to BOTH questions. Each question carries equal weight. You are advised to initially spend no more than 45 minutes on each question and then to return to any incomplete questions if you have time at the end. An indication of the number of marks allocated to parts of questions is shown in square brackets.

No calculator is permitted in this examination.

Section A

1. (a) (i) Give the definition of a linear map.
- (ii) Verify that the map $\phi : P_1 \rightarrow P_1$ given by $f \mapsto xf' + f(1)$ is a linear map. Here P_1 stands for the space of all polynomials of degree up to 1 and f' is the derivative of the polynomial $f \in P_1$. [5]
- (b) Verify that the matrix of ϕ with respect to the standard basis of P_1 is $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. [4]
- (c) (i) Which linear maps are called linear isomorphisms?
- (ii) Using T above, check whether the map ϕ from (a)(ii) is a linear isomorphism. [3]
- (d) (i) Define the kernel of an arbitrary linear map $\psi : U \rightarrow V$.
- (ii) Prove that ψ is injective if and only if $\ker(\psi) = 0$.
- (iii) Based on your answer in (c)(ii), what can you say about $\ker(\phi)$? Explain. [8]
- (e) (i) Find the eigenvalues and eigenvectors of the matrix T above.
- (ii) Is T diagonalisable? Explain. [5]

Section B

2. (a) (i) Give the definition of a bilinear form.
- (ii) Under which conditions is a bilinear form called an inner product? [5]
- (b) A symmetric bilinear form Φ on a 2-dimensional vector space U has the Gram matrix $\begin{pmatrix} a & b \\ c & 1 \end{pmatrix}$ with respect to a basis $\{u, v\}$. Find the entries a , b , and c if it is given that $\Phi(u, u) = 1$ and $\Phi(u + v, v) = 0$. [4]
- (c) Let V be an inner product space.
- (i) Which sets of vectors in V are called orthogonal?
- (ii) Prove that every orthogonal set of non-zero vectors is linearly independent. [6]
- (d) Consider the space P_2 of all polynomials of degree up to 2 as an inner product space with respect to the inner product

$$(f, g) = \int_{-1}^1 f(x)g(x)dx.$$

- (i) Verify that the polynomials 1 and x are orthogonal.
- (ii) Find the orthogonal projection of $3x^2$ to the plane $W = \langle 1, x \rangle$. Clearly state which polynomial is the projection. [5]
- (e) Consider $\mathbb{R}^4$ as an inner product space with respect to the dot product

$$u \cdot v = u_1v_1 + u_2v_2 + u_3v_3 + u_4v_4,$$

where $u = (u_1, u_2, u_3, u_4)$ and $v = (v_1, v_2, v_3, v_4)$. Use the Gram-Schmidt process to construct an orthogonal basis in the subspace $E = \langle (1, -1, 0, 0), (1, 1, 1, -1), (2, 0, 2, 0) \rangle$. [5]

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LI Linear Algebra

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats and outer-wear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) **MUST** be placed in the designated area.
- Check that you **DO NOT** have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **MUST** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are **NOT** permitted to use a mobile phone as a clock. If you have difficulty in seeing a clock, please alert an Invigilator.
- You are **NOT** permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately.
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to the Student Conduct procedures.